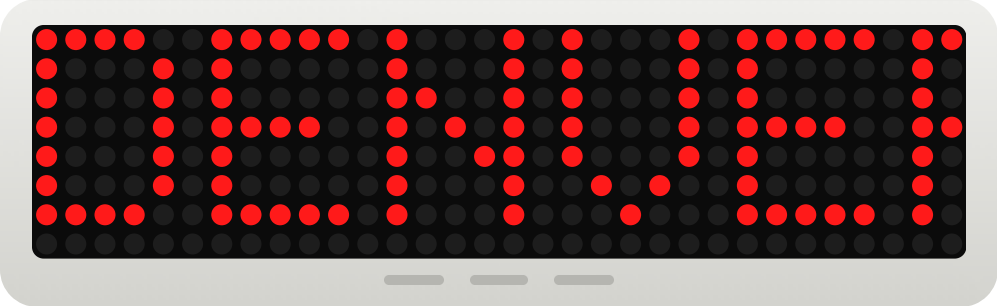
QUICK START GUIDE

Overhead

A pixel clock that shows where the plane over your house came from, or where it's going, only while you can actually hear it.



A Delta flight from Denver passing overhead. Between planes it's a clock.

What you need

Ulanzi TC001
The pixel clock (about $40–50). Nothing to open or solder.

USB-C data cable
For a one-time install from a computer. Some cables only charge.

Chrome or Edge
On a computer, for the installer. Then a phone for setup.

Four steps

1. **Install Overhead on the clock**
   Download `overhead-tc001.bin` from the project's Releases page on GitHub (github.com/MishaTikh/overhead-clock). Plug the clock into the computer, open `espressif.github.io/esptool-js` in Chrome or Edge, click Connect, pick the USB serial port, add the file at address `0x0` and click Program. Takes about a minute.

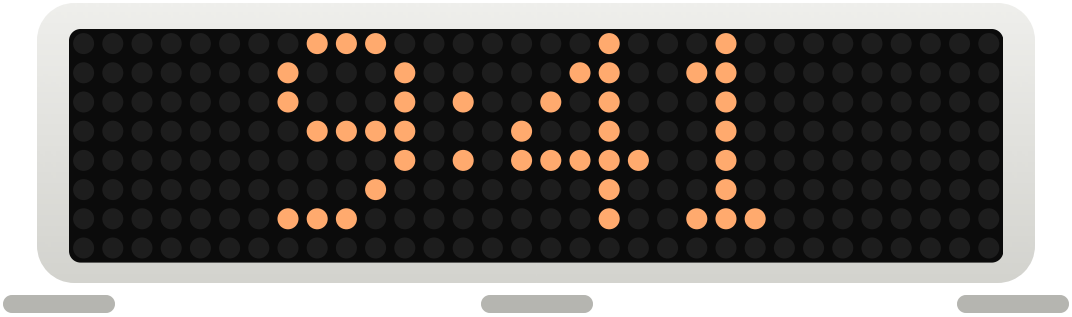
2. **Plug it in at home**
   The clock scrolls its setup Wi-Fi name and password, for example `WIFI OVERHEAD-1A2B PASSWORD K7PR3XQ9TMWA`.

3. **Connect it with your phone**
   Join that Wi-Fi network with the password shown. A setup page opens (or go to `192.168.4.1`). Pick your home Wi-Fi, type its password and your street address, then Save.

4. **Open the settings page**
   The clock scrolls its address, user and password. On a phone or computer on the same Wi-Fi, open `http://overhead.local` (or the number it shows), sign in as `admin` and check the address was found.

USING IT

Three buttons

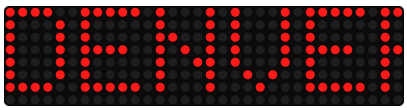
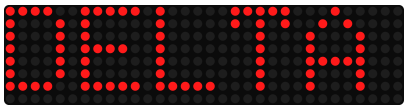
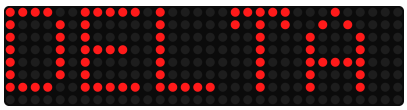


Left
Press: what to show for each plane (cycles the four styles below).
Hold: replay recent planes.
Held while plugging in: forget Wi-Fi.

Middle
Press: switch between *Clock + planes* and *Planes only*.
Hold: show the settings address and password.

Right
Press: brightness, cycling Auto, Low, Medium, High.
Auto follows the room light.

What it shows for each plane

City	Where a landing plane came from, or where a departing plane is going. The default.	
Codes	Airport codes, from and to.	
Airline	Airline name with the codes.	
Full	Everything: flight, cities and aircraft type.	

Long text scrolls. If a flight's route isn't known, every style shows the flight number. Colors follow the airline (Delta red, United blue, Sun Country orange…).

Two modes

Clock + planes (default)
Shows the time; switches to the flight while a plane is audible, then back.

Planes only
Stays dark until a plane is audible. Nice for a bedroom or a shelf you don't want glowing.

SETTINGS PAGE

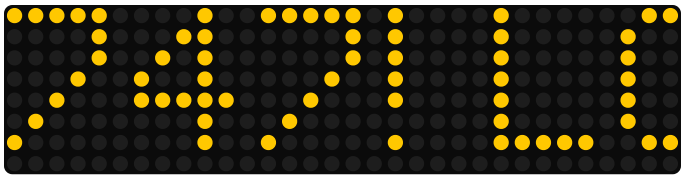
Everything else is on your phone

Open `http://overhead.local` (user `admin`). Changes save as you make them. Some phones don't open `.local` names: hold the middle button to see the clock's number instead.

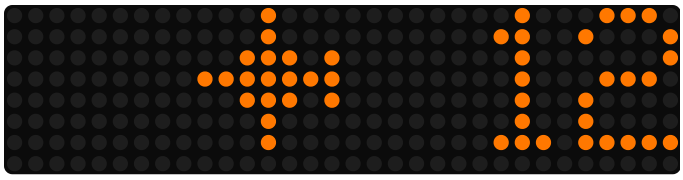
SECTION	WHAT YOU CAN DO
Today	Planes heard today, the farthest origin, the loudest plane, rare visitors, and an hour-by-hour chart against a usual day (learned over the first week).
Airport weather	Temperature and wind at your airport, and whether the wind has planes *landing over you*, *taking off over you* or going elsewhere today.
Home location	Street address (looked up once) or coordinates if you'd rather not send an address anywhere. Airport (nearest by default) and time zone (automatic).
What the sign shows	Mode, display style, 12 or 24-hour clock.
Alerts	Rare aircraft in gold with a sparkle; an optional chime; an inbound countdown with a little plane icon; an evening recap at the hour you choose.
Clock screen extras	Every so often, briefly show: indoor temperature and humidity, outdoor weather, date, sunrise and sunset, whether planes are over you today, planes heard today. All off by default.
Colors	Clock color, airline colors on or off, and a color for other planes.
Brightness and text	Auto, low, medium or high; dark-room and daylight levels; scroll speed; try any text on the sign.
Quiet hours	No plane alerts or data requests overnight (11 pm–6 am by default). The clock keeps showing the time.
Sensitivity	How loud a plane must be to show (60 dBA is clearly noticeable over a quiet street); daily data budget; saving requests when planes are elsewhere.
Wi-Fi and security	Network and signal, change the admin password, forget Wi-Fi.

Getting the sensitivity right

For the first few days, compare the *Recent* list with what you hear. If the sign lights before a plane is clearly audible, raise the threshold a few dBA; if it misses planes you notice, lower it.



A rare visitor: gold, with a sparkle.



Inbound countdown: a plane is 12 seconds from audible.

GOOD TO KNOW

Troubleshooting

IF…	TRY THIS
The installer can't find the clock	Use a different USB-C cable (some only charge) and Chrome or Edge. On Windows, the CH340 USB driver may be needed.
Your phone doesn't open the setup page	Stay on the clock's Wi-Fi and browse to `192.168.4.1`.
You changed Wi-Fi or moved	Hold the left button while plugging the clock in. It forgets the old network and shows the setup Wi-Fi again.
`overhead.local` doesn't open	Hold the middle button; open the number it scrolls, like `192.168.1.42`.
Forgot the admin password	Hold the middle button: it scrolls the password.
It shows flight numbers, not cities	The route for that flight isn't known yet. Common at airports without a supported flight board; add one for yours (see CONTRIBUTING on GitHub).
Nothing ever shows	Check the settings page: address found? Airport nearby? Inside quiet hours? Planes may be using other runways today (see Airport weather).
Wi-Fi or the internet is down	Nothing to do: it keeps working as a clock and picks up planes again when the connection returns.

Privacy and security

Stays at home
No account, cloud or remote access, and no internet port opened. The setup Wi-Fi and the settings page are password protected. Updates need a USB cable at the clock.

What leaves the house
Your address once (or nothing, if you enter coordinates), your location rounded to about 1 km when asking for nearby planes, and flight numbers when looking up routes. Every connection is verified HTTPS.

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